

Arsh Verma

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High agency ML Engineer with industry experience owning end-to-end, production-grade ML systems, from model development to real-world deployment, and stakeholder interactions. Skilled in aligning technical decisions with product goals for tangible impact.

Work Experience

Associate Machine Learning Scientist - 2 at Wadhvani AI

Aug 2021 – Aug 2024

- Deployed a production-grade deep learning system for detecting 17 chest X-ray abnormalities, improving mean AUROC to 90% (~3% improvement over SotA) and serving 2M+ patients annually on India's national digital diagnosis platform.
- Built and deployed a CXR-based TB screening model for the National Health Mission, aiding diagnosis in over 3M presumptive TB cases per year; achieved 90% sensitivity with 80% specificity on low-quality CXR photographs, exceeding WHO screening tool benchmarks.
- Enabled smartphone-based inference of CXR photos through 90% model compression via pruning and quantization.
- Earned a fast-tracked promotion for end-to-end project ownership, technical leadership, and mentoring 7 junior engineers.

Research

Learning Policies for Efficient and Coordinated Real-time Multi-Robot Search

Aug 2024 – Jul 2026

Advisor: Dr. Jeff Schneider (Research Professor at Carnegie Mellon University (CMU))

- Designed route-aware planning strategies for multi-robot Bayesian target search, and deployed them on real robots in a 75,000 m² field.
- Distilled expert planners into GNN policies, reducing on-robot inference time and completing missions 2.9× faster than SotA baselines.
- Developed a multi-agent Q-learning framework, enabling coordination between UAV & UGV teams using a shared belief formulation.
- Outperformed the strongest learned baseline for search efficiency (0.910 vs. 0.875 F-Score), with the margin scaling with team size.
- Demonstrated robustness to mid-mission robot failures and communication outages (up to 75% of the budget), all without retraining.

Publications

QUEST: Q-Learning for Uncertainty-Guided Efficient Search Teams Verma A.^{*}, Gupta T.^{*}, Schneider J. at *NeurIPS 2026 (under review)*

Efficient Active Search via Amortized Path-Integral Policies Verma A.^{*}, Gupta T.^{*}, Schneider J. at *ICRA 2026*

Generalized Cross-domain Multi-label Few-shot Learning for Chest X-rays Aimen A., Verma A., Tapaswi M., Krishnan N. C., at *ISBI 2025*

Using LLMs in Software Requirements Specifications: An Empirical Evaluation Krishna M., Gaur B., Verma A., Jalote P. at *RE Conference, 2024*

Towards long-tailed, multi-label disease classification from chest X-ray Holste, G. et al, in *In Medical Image Analysis, 2024*

How Can We Tame the Long-Tail of Chest X-ray Datasets? Verma A., at *ICCV-W: CVAMD, 2023*

Can we Adopt Self-supervised Pretraining for Chest X-Rays? Verma A. and Tapaswi, M., at *ML4H 2022*

Education

M.S. in Machine Learning and Robotics @ Carnegie Mellon University

CGPA: 4.24/4.00

Aug 2024 – July 2026

Research: Learning-based planning and decision-making for real-time robotic systems.

Coursework: Generative AI, DL Systems and Algorithms, Multi-Modal ML, Advanced Computer Vision, Reinforcement Learning.

B.Tech. in Computer Science @ IIT Delhi

CGPA: 9.04/10

Aug 2017 – Jun 2021

Coursework: Deep Learning, Advanced Machine Learning, Natural Language Processing, Probability & Statistics, Linear Algebra.

Projects

- RoPE + GQA for Transformers:** Implemented Rotary Positional Embeddings and Grouped-Query Attention in a Transformer, with Key-Value caching and causal masking. Benchmarked attention latency across sequence lengths and head configurations, reducing validation loss by >10% over a minGPT baseline and improving throughput/memory behavior.
- LoRA for Efficient LLM Fine-Tuning:** Implemented LoRA for GPT-2 by injecting trainable low-rank adapters into attention layers Achieved ~2% higher accuracy than full fine-tuning while updating <5% of parameters, benchmarking convergence and performance across multiple LoRA ranks.
- AI ChatBot for Census Data:** Developed an LLM-powered text-to-SQL agentic chatbot over US Census data, supporting multi-part queries, multi-turn context retention, and reasoning over structured data. Implemented guardrails for prompt injection, off-topic detection, and hallucination prevention.
- Needle: A Minimal Deep Learning Framework (Mini-PyTorch):** Built a DL framework from scratch. Supported dynamic computation graphs and reverse-mode automatic differentiation, core neural architectures (CNNs, RNNs, Transformers), optimizers, and GPU acceleration for end-to-end training.
- Diffusion Models (DDPM):** Built a DDPM with cosine noise scheduling and a U-Net-based noise predictor, implementing both forward noising and reverse denoising processes. Trained and sampled images via iterative reverse diffusion and achieving FID <160 after ~2 hours of training on a T4 GPU.

Interests & Skills

Research Interests: Deep Learning for Computer Vision, Language Modelling, Reinforcement Learning, Multi-modal Learning, AI for Healthcare.

Skills and tools: Python, PyTorch, Generative AI, Large Language Models (LLMs), Distributed Training, Parallelization, Docker, TorchServe, MongoDB.

Positions of Responsibility

Reviewer, Machine Learning for Health Conference

[2023, 2024]

Head Teaching Assistant for the graduate-level ML course. Managed a team of 7 TAs for the course taken by 150 UG, Masters and PhDs.

[2020, 2021]

Member, Chess Club, Math Club and Poker Club @ IIITD and CMU

[2017-2025]